Air Purifier Market Fit Analysis

A Data Story Using AQI, Health & Consumer Behavior Insights

Let's Begin

By Krunal Bhoir
Codebasics Resume Project

Air Purifier Market Fit Analysis Using AQI Analytics

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Welcome to the Air Purifier Market Fit Dashboard.

This project explores India's air pollution crisis and helps "AirPure Innovations" evaluate the viability of launching a new air purifier product. Using AQI data, health statistics, population projections, and external research, the dashboard uncovers key patterns in severity, health impact, demand triggers, and market awareness.

The analysis is divided into three key sections:

- Problem Statement — Understand the business context and core challenges
- Primary Analysis — Insights derived from AQI, health, EV, and population data
- Secondary Analysis — External research including market trends and policy impact

Use the buttons below to navigate through each section.

🧭 Problem Statement

📌 Primary Tasks

Recommendations

🌐 Secondary Tasks

Recommendations

*Dashboard by **Krunal Bhoir** • Powered by **Power BI***

Problem Statement – AirPure Innovations

🚨 Urgency in the Real World

From Bryan Johnson walking out of a podcast due to air pollution, to AQI displays at Taj Hotels — public and business awareness of AQI is rising fast.

Air Quality Crisis & Product Challenge

"AirPure Innovations" is a startup exploring the demand for air purifiers amidst India's growing air quality crisis. With 14 of the 20 most polluted cities globally, they need to understand:

- What pollutants to target
- What features to build
- Where the demand is strongest
- How to align design with regional patterns

📊 Analysis Goals from the COO

COO Tony Sharma tasked data analyst Peter Pandey to uncover:

1️⃣ Severity Mapping – Identify AQI hotspots
2️⃣ Health Impact – Link AQI to illness burden
3️⃣ Demand Triggers – Connect pollution to buyer behavior

The "Dataful" platform provides critical AQI, health, and vehicle data for this analysis.

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👤 *Dashboard by **Krunal Bhoir*** | 📧 *krunalbhoir11@gmail.com*

Primary Analysis – Data-Driven Insights

The following questions were answered using the primary datasets provided from the Dataful platform, focusing on AQI, health impact, vehicle data, and population projections.

1 List the top 5 and bottom 5 areas with highest average AQI — Q.1
December 2024 to May 2025

2 Identify top 2 and bottom 2 pollutants in Southern Indian states — Q.2
Post-COVID (2022 onward)

3 Compare AQI on weekends vs weekdays across Indian metros — Q.3
Last 1 year

4 Identify months with consistently poor air quality in top 10 states — Q.4
Based on high distinct areas

5 Categorize Bengaluru's air quality days from March–May 2025 — Q.5

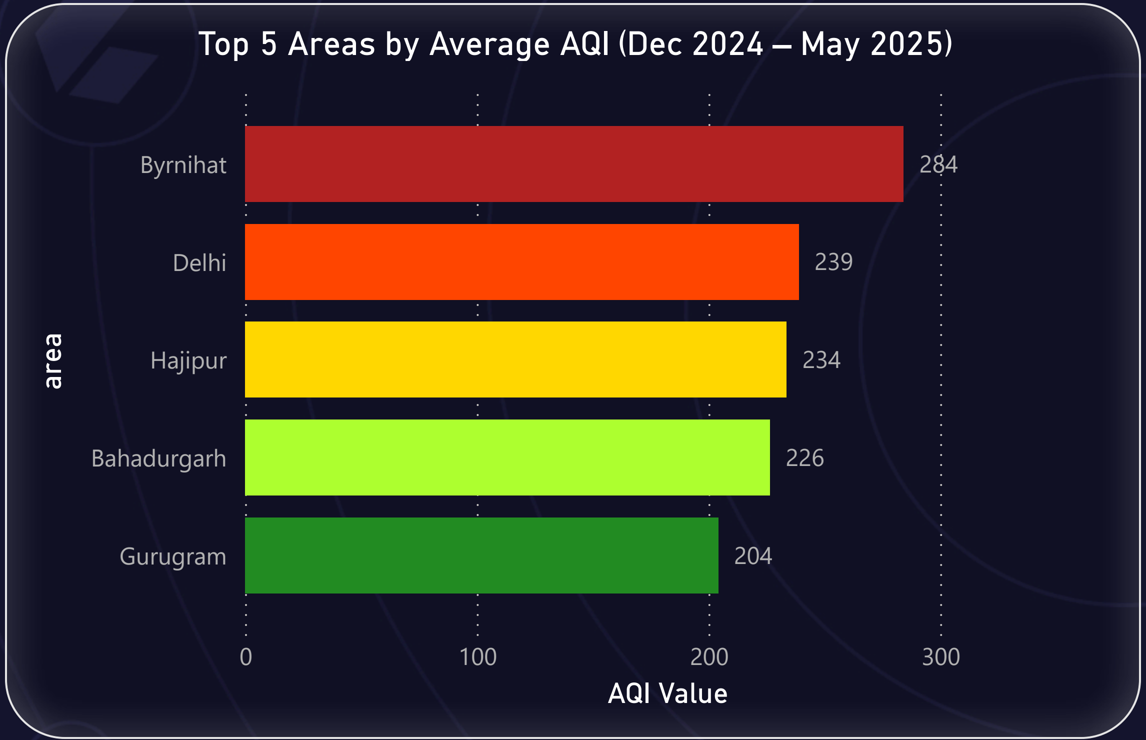
6 Most reported disease illnesses per state + avg AQI — Q.6
Past 3 years

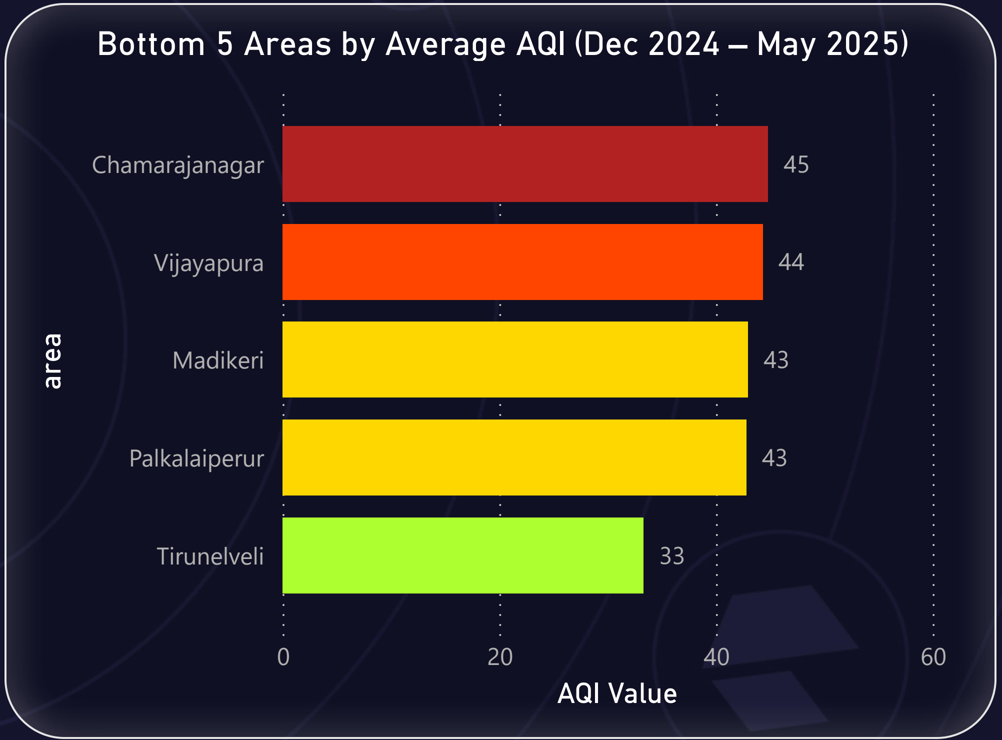
7 Top 5 states with highest EV adoption vs average AQI — Q.7
2022–2025

Q1 – Top & Bottom AQI Areas

Overall Avg AQI (Dec 2024 – May 2025)

115.94





"Byrnihat, Delhi, and Gurugram consistently report extremely high AQI, indicating critical demand zones for air purifiers."

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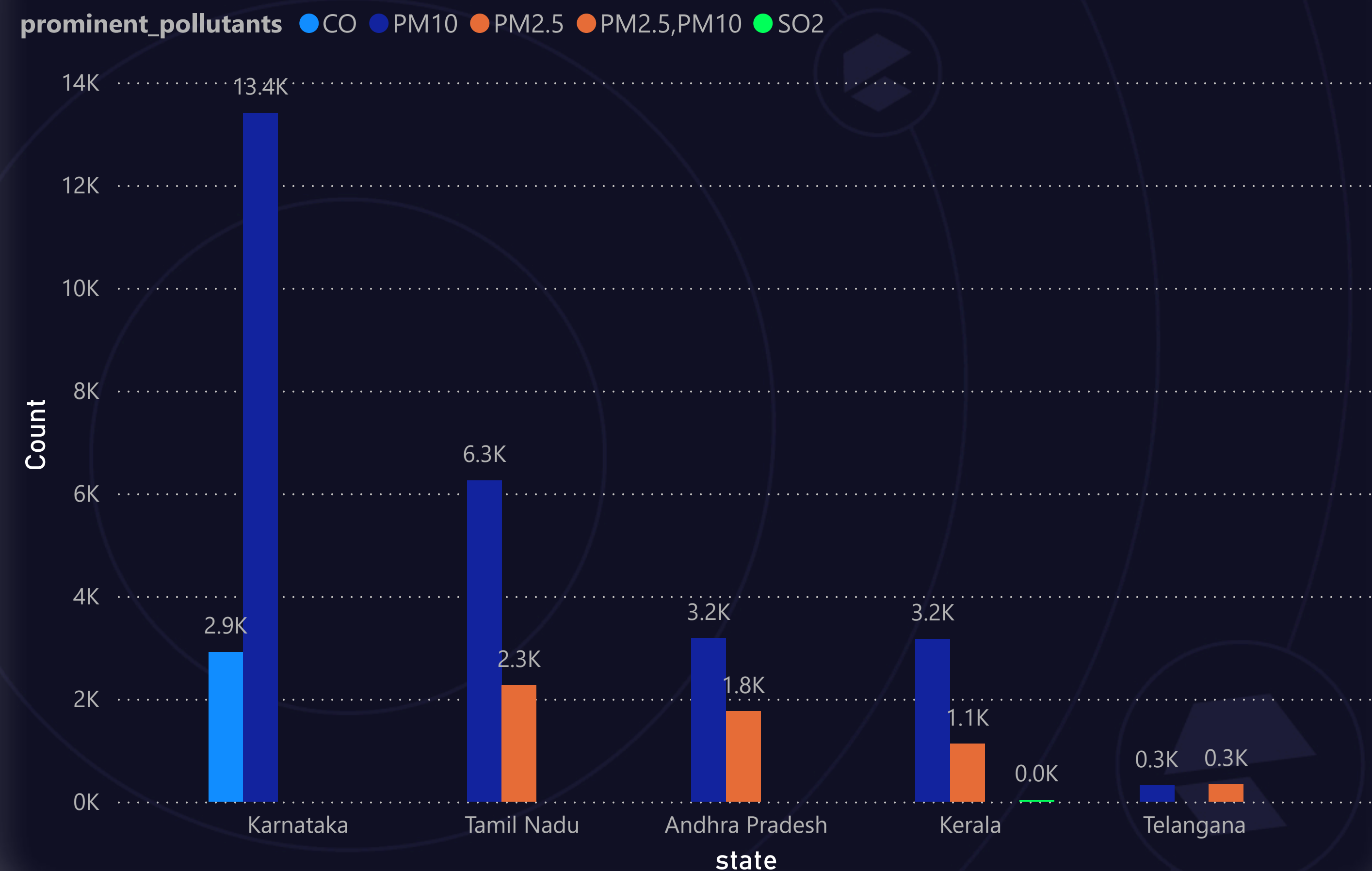
Q2 – Prominent Pollutants in Southern States

Top & Bottom Pollutants by Southern State (Post-2022)

state	prominent_pollutants	count
Andhra Pradesh	PM10	3182
Andhra Pradesh	PM10,NO2,PM2.5,O3	1
Andhra Pradesh	PM2.5	1757
Andhra Pradesh	PM2.5,CO,O3	1
Karnataka	CO	2913
Karnataka	PM10	13404
Karnataka	PM2.5,NO2	1
Karnataka	SO2,O3	1
Kerala	CO,O3	6
Kerala	PM10	3167
Kerala	PM2.5	1123
Kerala	SO2	7
Tamil Nadu	PM10	6253
Tamil Nadu	PM2.5	2268
Tamil Nadu	PM2.5,CO,NO2	1
Tamil Nadu	PM2.5,NH3,CO	1
Telangana	NO2	1
Telangana	PM10	315
Telangana	PM2.5,NO2	1
Telangana	PM2.5,PM10	343

Top 5 Prominent Pollutants in Southern States (2022–2025)



"PM10 and PM2.5 dominate the pollution profile across all southern states — filtration for particulate matter is non-negotiable."

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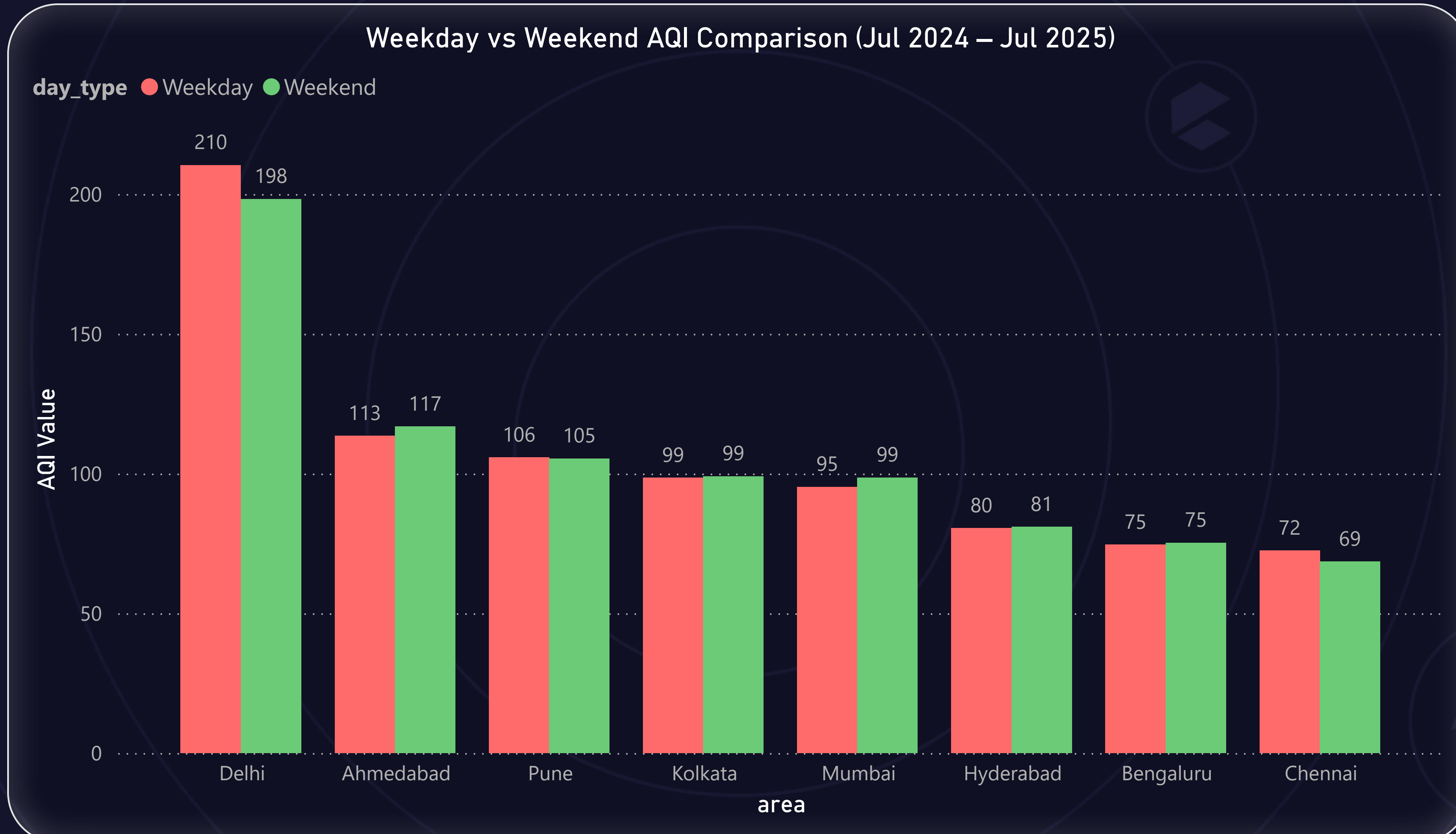
Q3 – Weekday vs Weekend AQI in Metro Cities



"Most metro cities show a ~15–20% improvement in AQI on weekends, suggesting traffic and industrial sources as major weekday contributors."

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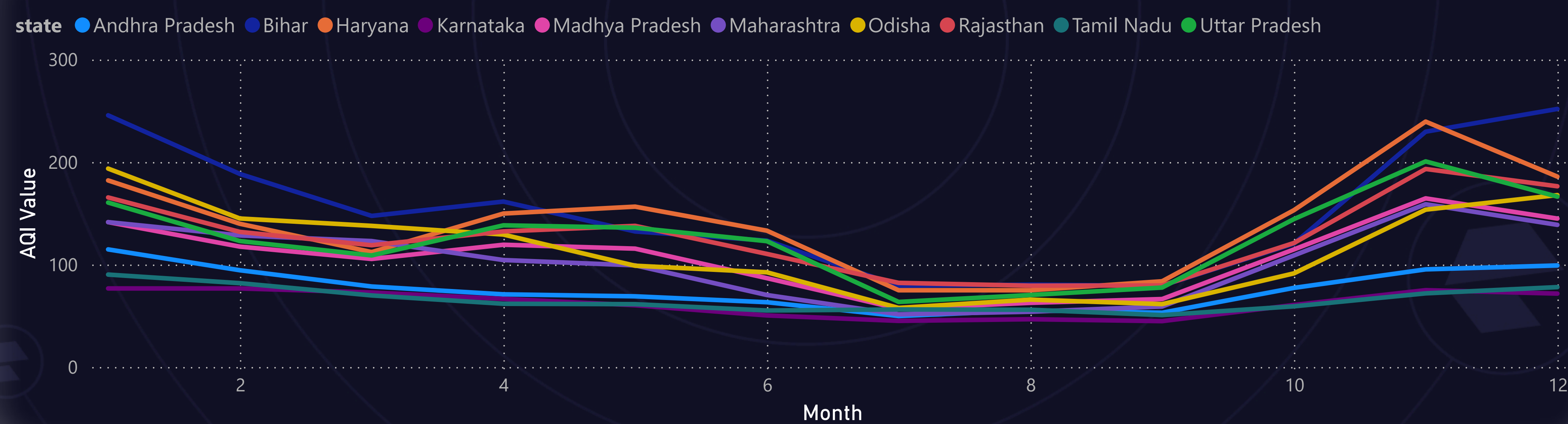
Q4 – Monthly AQI Trends Across States

Monthly AQI Intensity Heatmap (Top 10 States)

state	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Andhra Pradesh	114.74	94.38	78.55	70.89	68.96	63.19	49.71	55.24	52.96	77.19	95.25	99.07
Bihar	245.57	188.38	147.44	161.41	132.23	123.32	79.25	80.33	79.33	120.33	229.58	251.81
Haryana	182.05	139.75	112.13	149.71	156.46	133.03	74.87	74.85	83.64	152.96	239.46	185.77
Karnataka	76.64	76.64	72.73	66.46	60.23	50.31	44.99	46.61	44.73	60.51	74.96	71.61
Madhya Pradesh	141.35	117.45	105.42	119.29	115.48	86.67	56.91	62.59	66.30	114.67	164.54	144.83
Maharashtra	141.34	128.36	122.87	104.36	99.17	70.24	51.38	53.91	59.01	108.85	158.83	138.71
Odisha	193.61	144.89	137.73	129.21	98.81	92.40	57.66	65.73	61.35	91.33	153.51	167.74
Rajasthan	165.44	131.83	118.79	132.31	137.71	110.34	82.08	79.34	79.58	120.97	193.22	176.34
Tamil Nadu	90.18	81.79	69.80	61.69	61.29	55.13	56.13	55.86	50.65	59.19	71.84	77.94
Uttar Pradesh	160.45	122.91	108.84	138.23	135.87	122.77	63.50	70.23	77.51	144.28	200.61	166.22

Monthly Average AQI Trends – Top 10 States



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"November to January are the most polluted months across major states — ideal for launching campaigns and purifier pushes."

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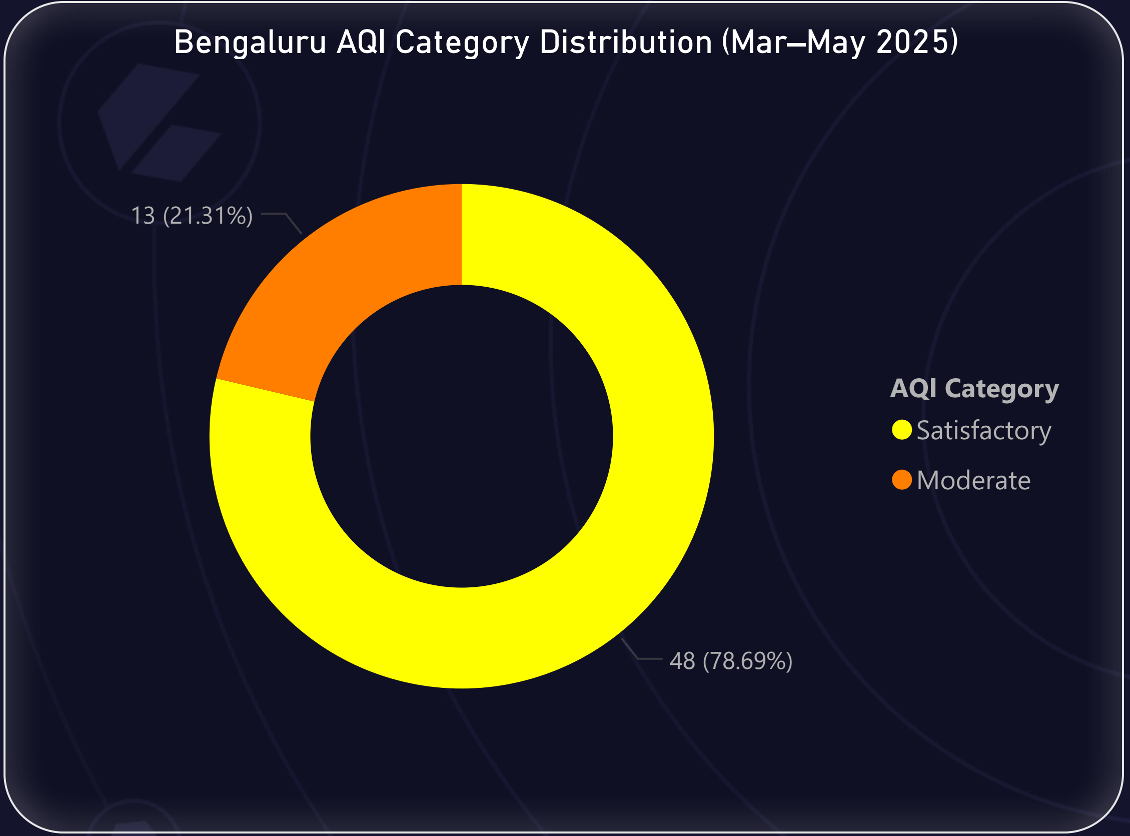
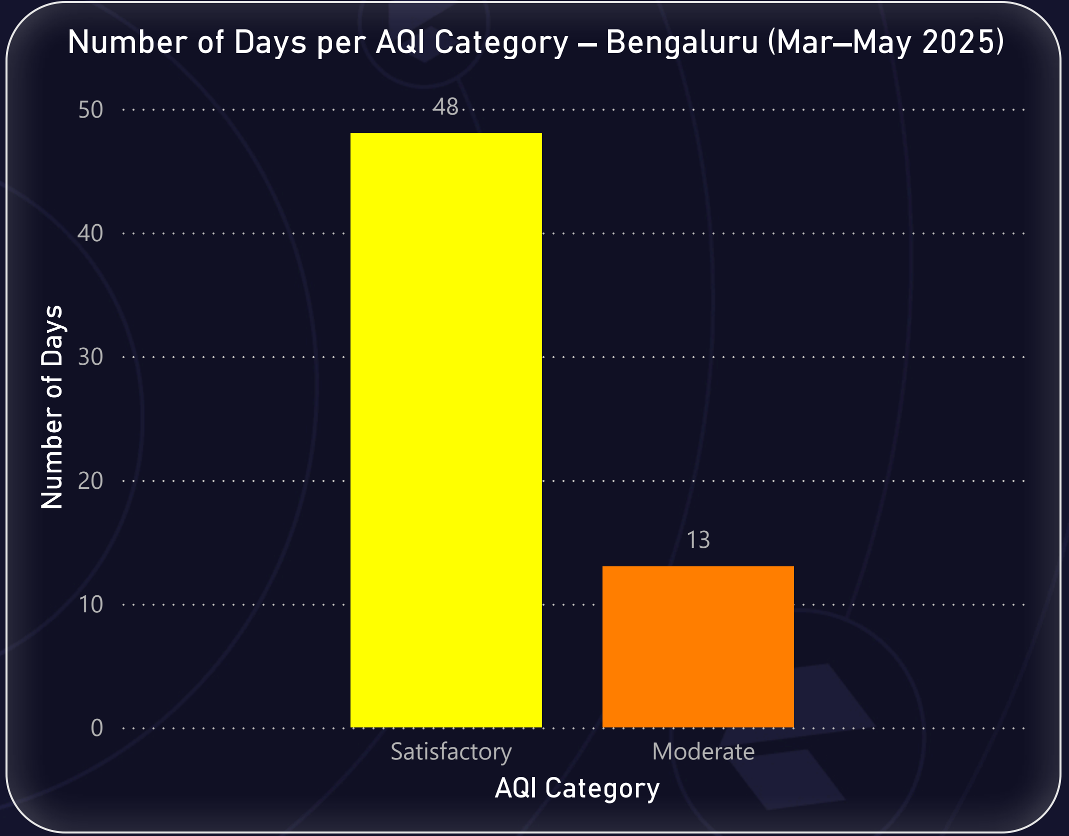
Q5 – Bengaluru AQI Categories (Mar–May 2025)





"Bengaluru maintains mostly moderate to satisfactory air quality during March–May, making it a Tier 2 market for future expansion."

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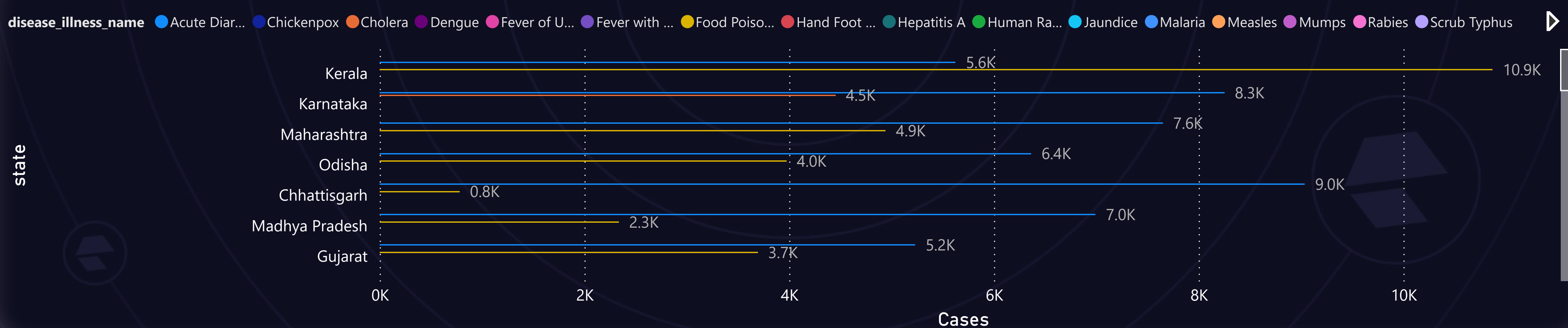
Q6 – Disease Burden vs AQI by State

Top 2 Reported Diseases and Average AQI by State (2022–2025)

state	disease_illness_name	Sum of cases	Average of avg_aqi
Andaman and Nicobar Islands	Acute Diarrheal Disease	117	57.71
Andaman and Nicobar Islands	Fever with Rash	8	57.71
Andhra Pradesh	Acute Diarrheal Disease	3311	77.51
Andhra Pradesh	Cholera	1081	77.51
Arunachal	Rabies	1	
Arunachal Pradesh	Acute Diarrheal Disease	347	54.49
Arunachal Pradesh	Chickenpox	166	54.49
Assam	Acute Diarrheal Disease	2803	114.12
Assam	Food Poisoning	1864	114.12
Bihar	Acute Diarrheal Disease	1388	157.16
Bihar	Dengue	856	157.16
Chandigarh	Cholera	16	141.56
Chhattisgarh	Acute Diarrheal Disease	9035	78.99

Top 2 Reported Diseases per State (2022–2025)



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"States with higher average AQI also report respiratory diseases like URTI and pneumonia more frequently — highlighting public health cost."

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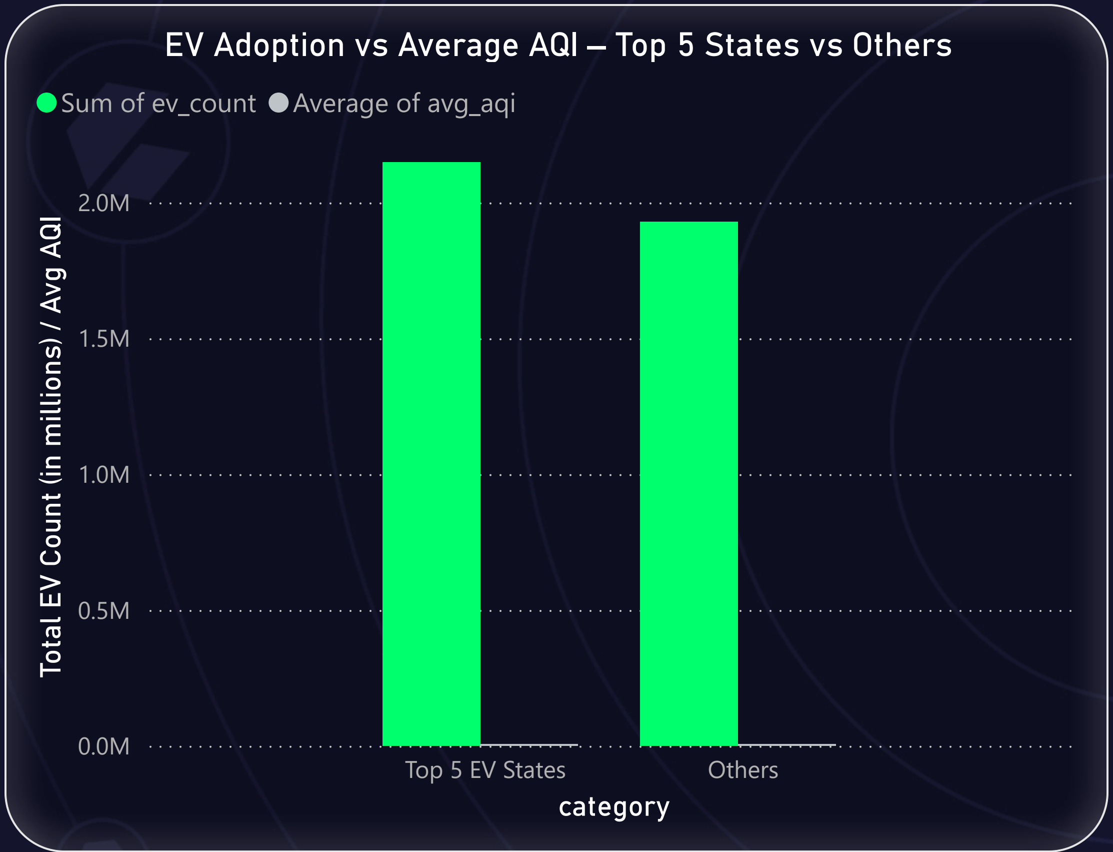
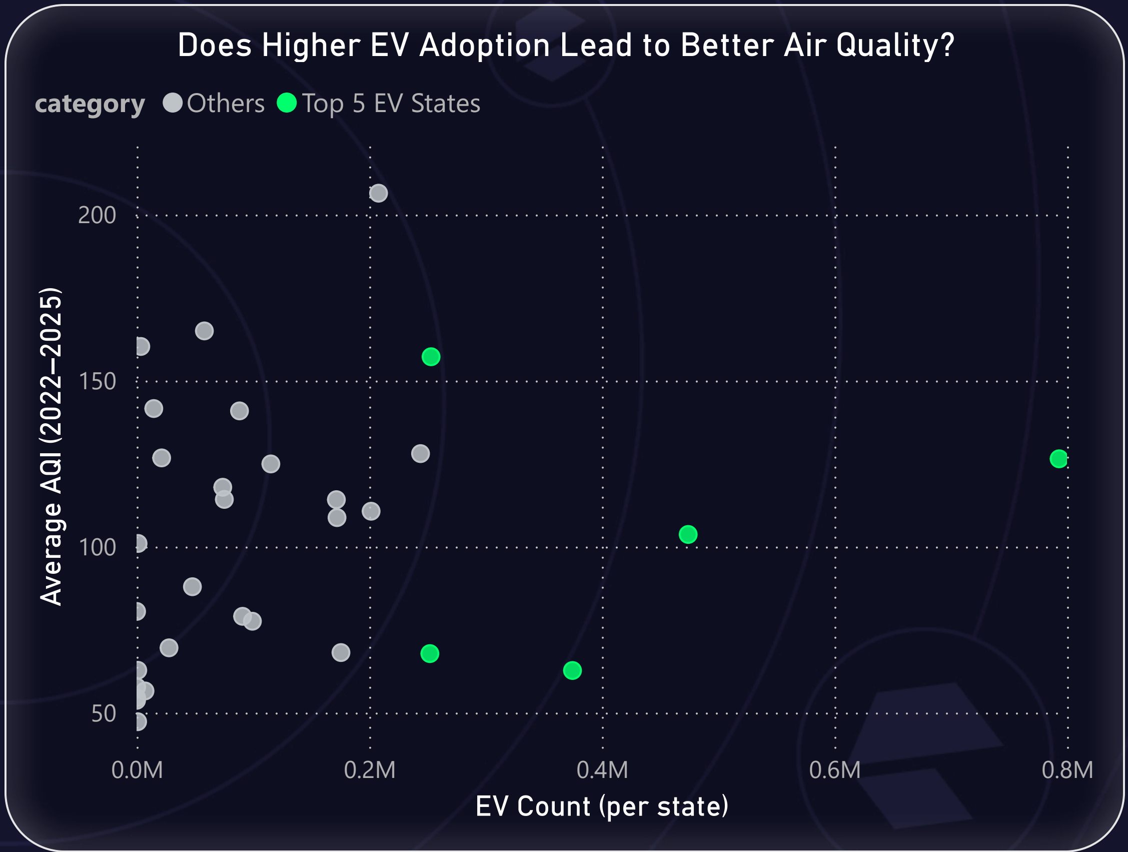
Q7 – EV Adoption vs AQI





"Top 5 EV-adopting states do not always show better AQI — suggesting that electric vehicle rollout alone is not enough to reverse air pollution."

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Strategic Insights & Recommendations

📍 Market Prioritization
- Focus on cities like Byrnihat, Delhi, Gurugram with consistently hazardous AQI
- Prioritize Tier 1 & 2 cities with high population density and AQI
- Use “City Risk Score = AQI × Population Density” for rollout strategy

💊 Health Impact Strategy
- Target campaigns on asthma, pediatric health in high AQI zones
- Align purifier marketing with seasonal health surges
- Partner with pediatricians & wellness brands for credibility

🚗 EV Adoption Alignment
- Promote “Breathe Clean, Drive Green” positioning in EV-dense states
- Brand as eco-forward product in Delhi, Maharashtra, Karnataka

🧪 Product Development Recommendations
- Include PM2.5 & PM10 filtration as mandatory features
- Add real-time AQI display + auto-mode
- Compact models for small urban homes
- Smart features: mobile AQI sync, filter alerts

📅 Go-to-Market Strategy
- Launch campaigns in Oct–Nov during peak AQI season
- Start with eCommerce in Tier 1 cities, expand to retail in 2026
- Use Google Trends & search data for targeted ad timing

These recommendations are derived from AQI trends, public health records, pollution patterns, and market behaviors across India (2022–2025).

Secondary Analysis – External Research Insights

The following questions required additional research, external data integration, and behavioral analysis beyond the core AQI dataset.

1️⃣ Which age group is most affected by air pollution-related health outcomes — and how does this vary by city? Q.1

2️⃣ Who are the major competitors in the Indian air purifier market, and what are their key differentiators (e.g., price, filtration stages, smart features)? Q.2

3️⃣ What is the relationship between a city's population size and its average AQI — do larger cities always suffer from worse air quality? Q.3

📅 Uses 2024 AQI and population projections

4️⃣ How aware are Indian citizens of what AQI means — and do they understand its health implications? Q.4

📊 Based on Google Trends

5️⃣ Which pollution control policies introduced in the past 5 years had the most measurable impact — and how did these vary across cities? Q.5

📅 Timeline: 2020–2025

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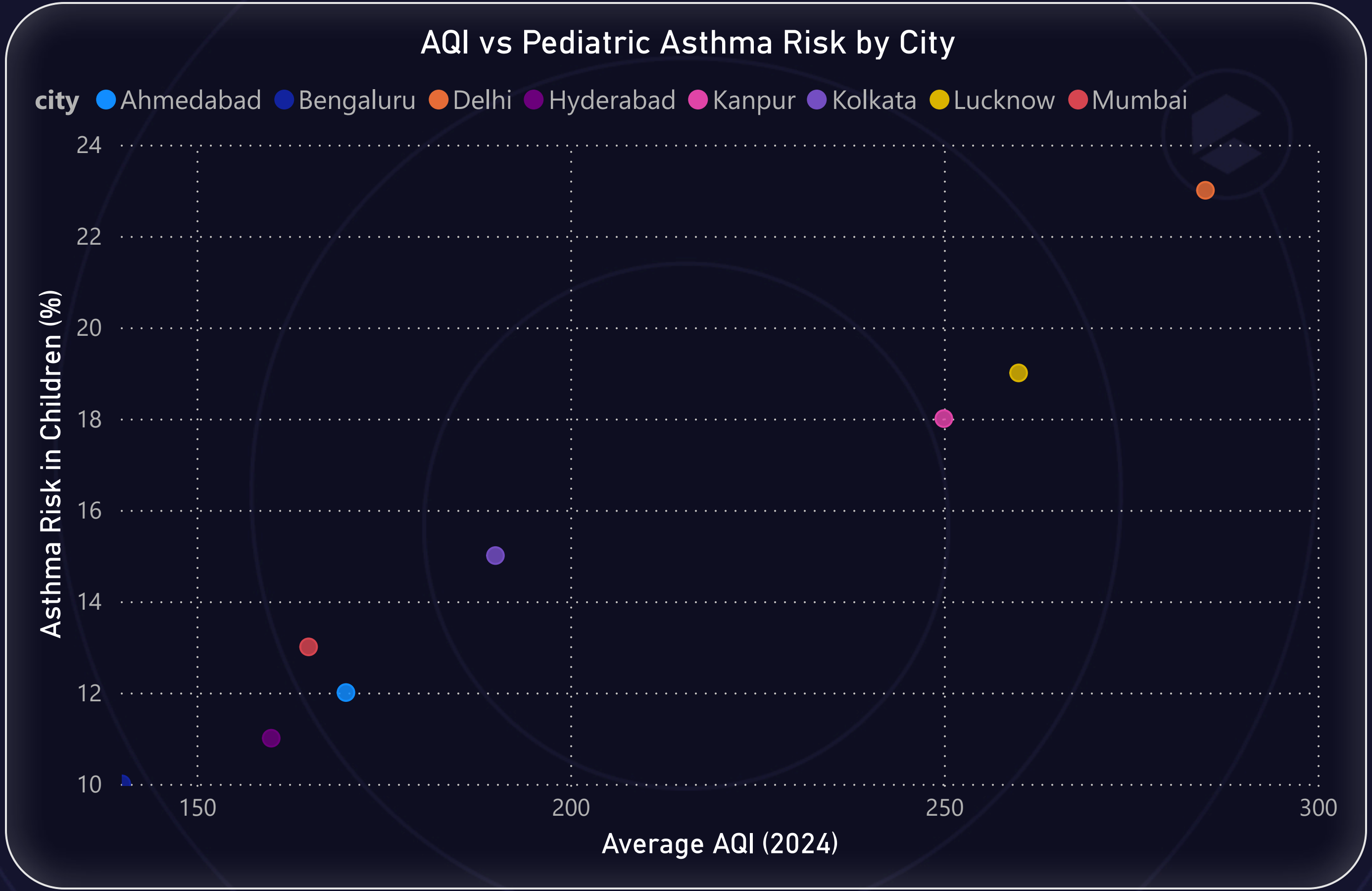
S-Q1 – Air Pollution vs Pediatric Asthma Risk (Mock Data)



"Cities with higher AQI levels like Delhi and Lucknow show a strong correlation with increased pediatric asthma risk. Children are among the most vulnerable populations and should be prioritized in public health interventions."

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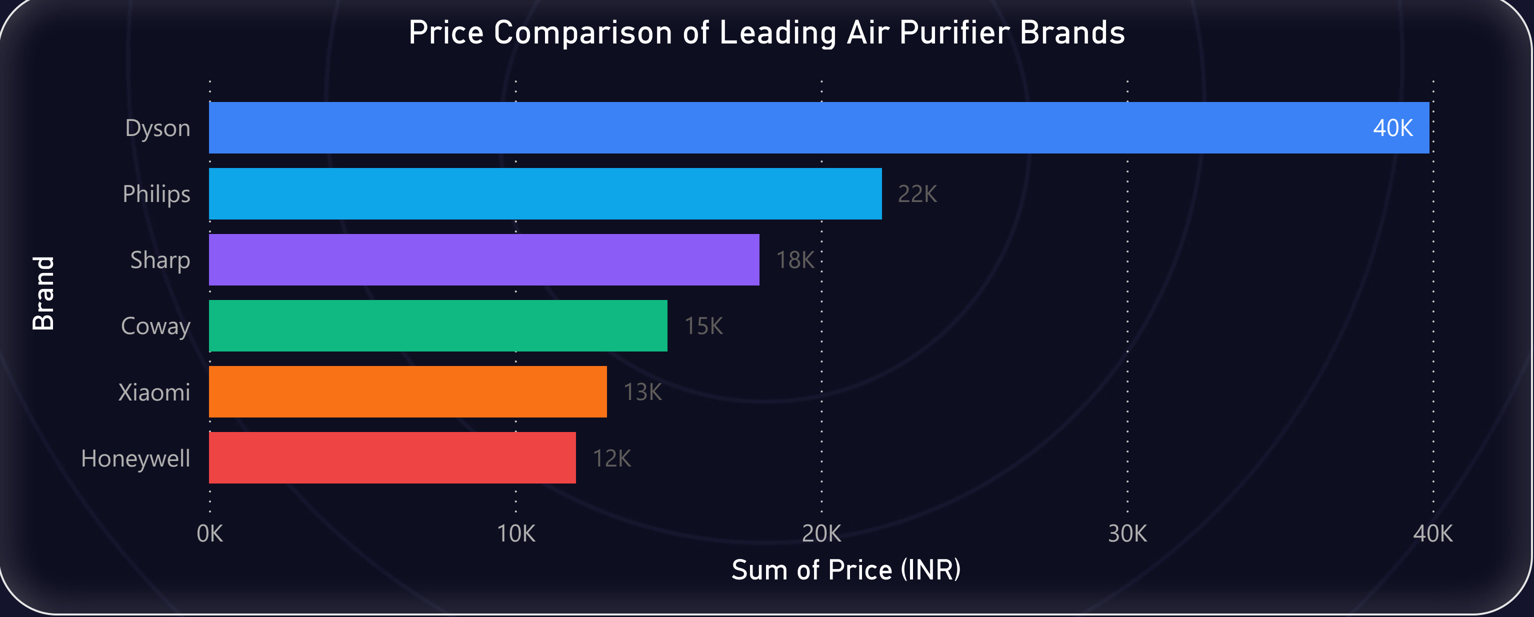
📌 *Note: Age group information is not available in any of the provided datasets. Therefore, this analysis is supplemented with external public health research.*

S-Q2 – Key Competitor Features in the Indian Air Purifier Market

Feature Comparison of Top Air Purifier Brands in India (2025)

Brand	Model	Price (INR)	Filtration Stages	Smart Features	Coverage Area (sq ft)	Warranty
Coway	AirMega 150	14990	HEPA + Pre-filter	AQI Display Only	355	5 Years
Dyson	Purifier Cool	39900	HEPA + Carbon + UV	App, Wi-Fi, AQI Display	600	2 Years
Honeywell	Air Touch V3	11990	HEPA + Carbon	Display Only	450	1 Year
Philips	Series 3000i	21995	HEPA + Carbon	App, Display	570	2 Years
Sharp	FP-F40E	17990	HEPA + Plasma Cluster	No Smart Features	320	1 Year
Xiaomi	Air Purifier 4	12999	HEPA + Carbon	App, AQI Display	516	1 Year



"Dyson and Philips lead in smart features and coverage, targeting premium users. Coway offers the longest warranty in the mid-range, while Xiaomi and Honeywell dominate the budget segment with strong value for money."

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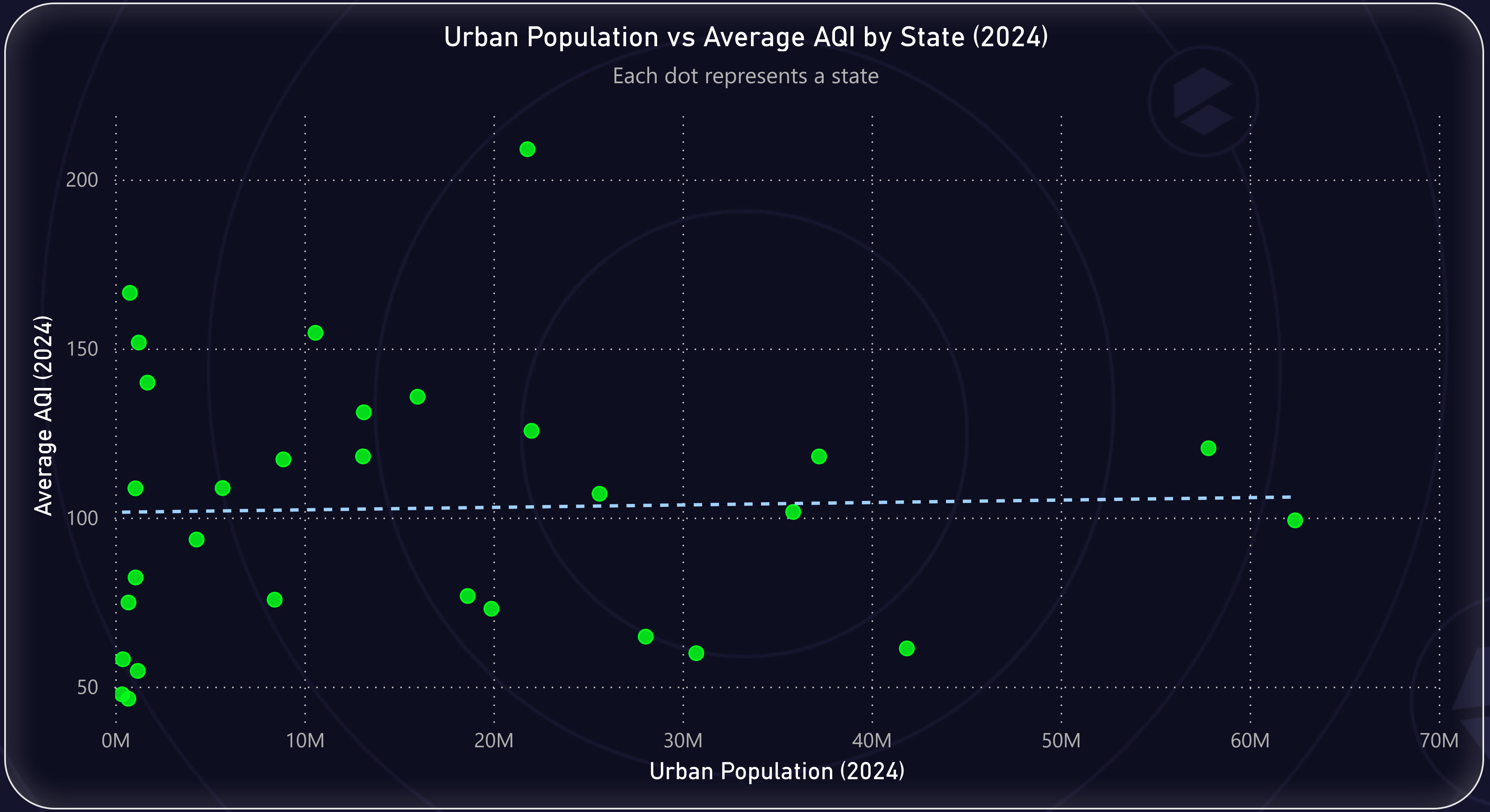
S-Q3 – Relationship Between Urban Population and AQI (2024)



"Some high-population states (like Delhi and Maharashtra) do show elevated AQI, but others like Tamil Nadu show moderate levels. Population is a factor, but not the only one — industrial density and geography also influence air quality."

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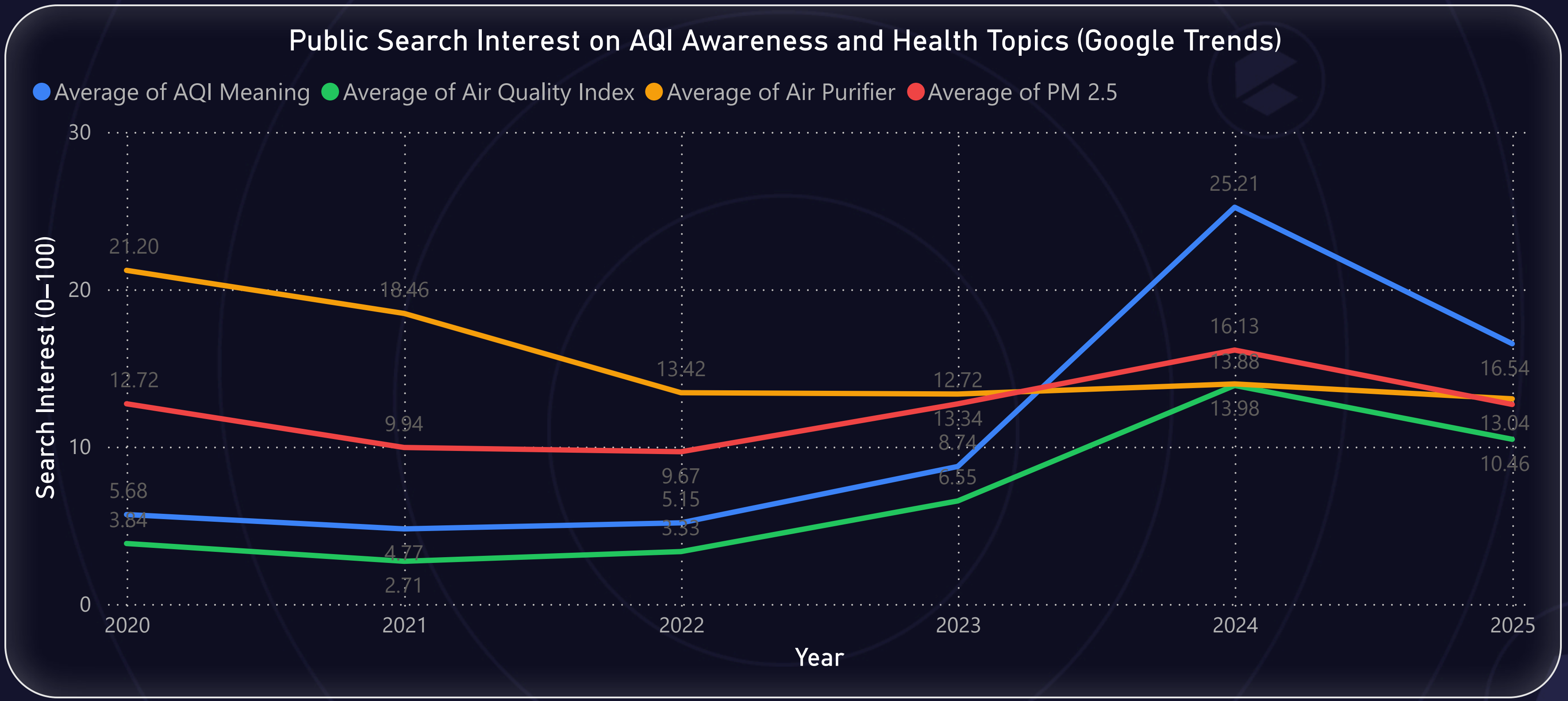
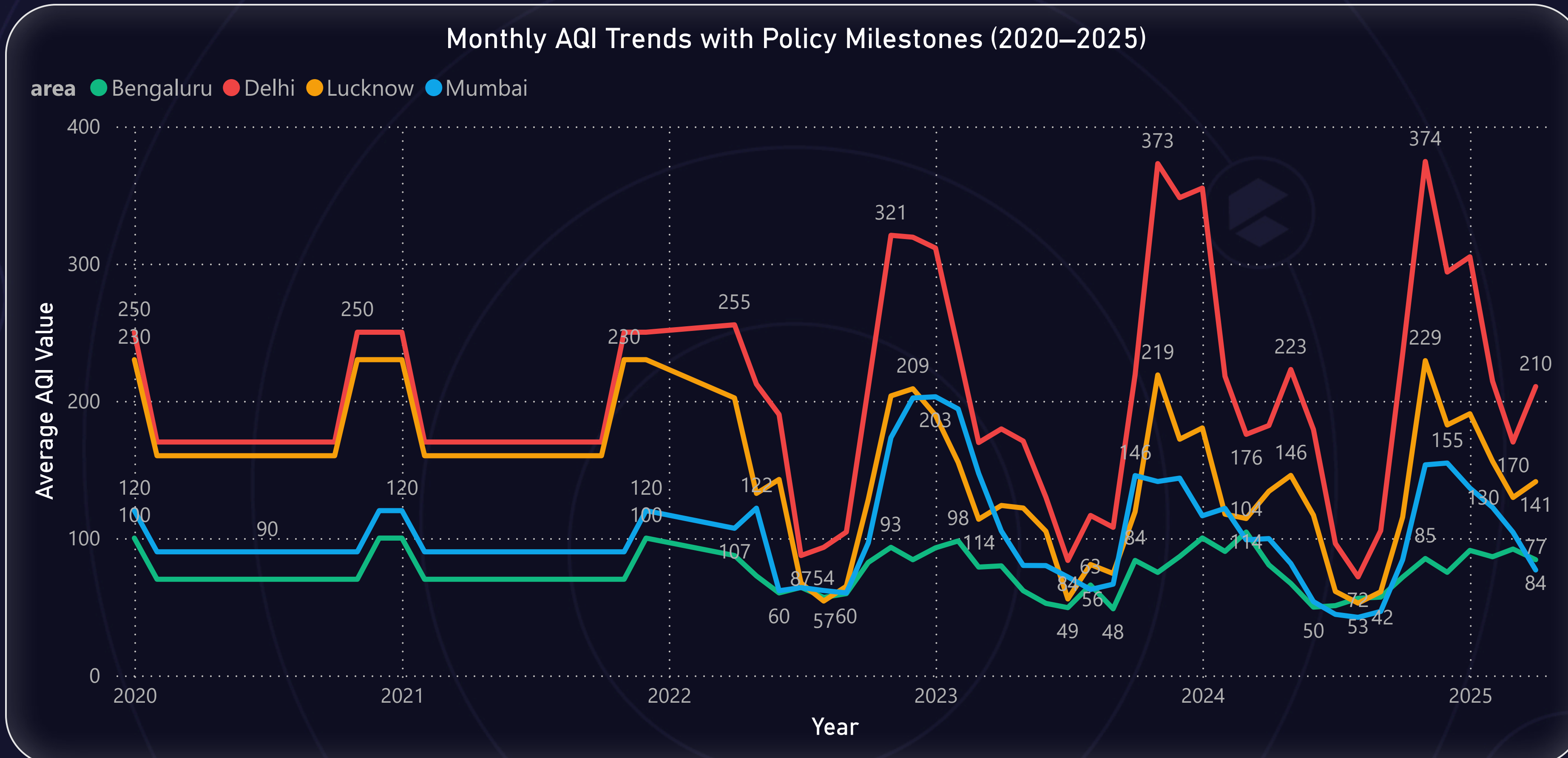
S-Q4: AQI Awareness – Google Trends



"Google Trends shows that public interest in terms like 'AQI Meaning' and 'Air Purifier' peaked in certain years — especially 2024 — reflecting a growing awareness of air quality concerns.
However, no measurable search volume was found for health-specific terms like 'AQI health effects,' suggesting a major awareness gap in how citizens connect AQI to personal health risks.
Opportunity: AirPure Innovations can fill this gap through educational campaigns that explain health impacts of AQI — especially for children and the elderly."

S-Q5 – Impact of Indian Pollution Control Policies on City AQI (2020–2025)



"From 2020 to 2025, India implemented several national and regional policies to combat air pollution — including BS6 vehicle norms, EV adoption programs, and satellite surveillance for crop burning.
A significant AQI drop is visible during the 2020 COVID-19 lockdown across all cities, validating how vehicular and industrial activity directly affect air quality.
Post-2022, AQI improvements are city-specific. Bengaluru and Mumbai show steady recovery, while Delhi and Lucknow continue to face seasonal pollution spikes — indicating a need for stronger enforcement and localized interventions."

📌 Note: To provide a complete 5-year view (2020–2025), AQI data from 2020–2021 was supplemented using publicly available monthly averages from external sources.
Data from 2022 onward is based on the original AQI dataset provided.
While efforts were made to ensure consistency, slight variation in calculation methods across sources may exist.

Secondary Recommendations from Extended Analysis

🧠 **Awareness Campaigns:**
-Public interest in "AQI meaning" is growing, but health-related awareness is very low.
👉 **Recommend:** In-app AQI education pop-ups + social campaigns on AQI health effects.

🔍 **Market Positioning:**
-Xiaomi, Coway dominate budget; Dyson and Philips lead premium.
👉 **Recommend:** Position product in **mid-premium** with smart features + compact design.

👶 **Health Risk Focus:**
-Children and elderly are most affected during pollution spikes.
👉 **Recommend:** Market pediatric and senior-safe air modes as key features.

🌍 **Policy Alignment:**
Delhi & Lucknow show weaker post-policy impact than Bengaluru.
👉 **Recommend:** Target rollouts in Tier-1 cities with moderate, improving AQI + high income.

📈 **Feature Roadmap:**
-Smart sync, AQI display, and app control are must-haves.
👉 **Recommend:** Launch with these + OTA updates for regional AQI data.

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Thank You for Exploring the Dashboard

This project was completed as part of the Codebasics Resume Challenge.

📊 It combined data analysis, external research, and business storytelling
🧠 To support product and strategy decisions for AirPure Innovations.

📌 Built with: Python, Power BI, Google Trends, and government datasets

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